

Standard candle: well-known intrinsic luminosity
Cepheid variables

• Type Ia Supernova

• RR Lyrae

$$m - M = 5 \log(D) - 5 + A$$

Standard ruler: well known intrinsic size

$$\frac{D_1}{\theta_1} = \frac{D_2}{\theta_2}$$

Absolute distance estimator: trig. parallax

rel. dist. estimator:

* MW:

• Trig. parallax: $\tan(\theta) \approx \theta$

$$1 \text{ pc} = 3 \times 10^{16} \text{ m}$$

$$D(\text{pc}) \approx \frac{1}{\theta}$$

* Spectroscopic data:

Cluster Main sequence fitting:

* Cepheid Variable stars: (Younger)

• Helium core $L \leq 100 L_\odot$

$$-2 < M_V < -6$$

• period: 2-100 days

$$M_I = -2.962 (\log(P) - 2) - 4.404$$

* RR-Lyrae Variables: older stellar populations

Luminosity methods:

- Tulley Fisher

- Faber - Jackson

→ Find V_{max} from HI rotational curve

- $L_{absolute} \propto V^2$

- Use L_{obs} to find Distance D
 $m - M = 5 \log(\underline{D}) - 5 + A$

Supernova:

Type I:

- white dwarf binaries

Type II: core collapse

Type Ia SNe:

- $\theta = \frac{R}{D}$: $\frac{d\theta}{dt} = \frac{\frac{dR}{dt}}{D} = \frac{V_{exp}}{D}$

V_{exp} velocity

$$D = \frac{V_{exp}}{d\theta/dt}$$

$$* L(t) = 4\pi R(t)^2 \sigma_{SB} T(t)^4$$

- Flux: $F(t) = \frac{L(t)}{4\pi D^2} = \underbrace{\left(\frac{R(t)^2}{D^2}\right)}_{\theta^2} \sigma_{SB} T(t)^4$

$$\underline{F(t)} = \theta^2$$

1/2

$$\sigma_{SB} T(t)^4 \quad \theta(t) = \frac{F}{\sigma_{SB}^{1/2} T^2}$$

Hubble: $V_{rad} = V_0 D + V_{pec}$

Galaxy Distribution:

- Integral # counts: total # of gal. brighter than flux density F

$$N(>F) \propto F^{-3/2}$$

Diff # count: # of galaxies per unit flux density at F

$$\tilde{N}(F) = \frac{dN}{dF} \propto F^{-5/2}$$

* Fixed L , $\phi \equiv \frac{N_{gal}}{Vol}$

$$F = L / 4\pi D^2$$

- Total # of gal. out to D :

$$N(<D) = \frac{4\pi}{3} D^3 \phi$$

all of these galaxies will have a Flux $F > \frac{L}{4\pi D^2}$

$$F_{lim} = \frac{L}{4\pi D^2} \Rightarrow$$

$$D = \sqrt{\frac{L}{4\pi F_{lim}}}$$

$$1 / 3/2 \cdot 4\pi$$

$$N(>F_{lim}) = \left(\frac{\Omega}{4\pi F_{lim}} \right) \frac{4\pi}{3} \phi$$

lit very faint galaxies typically lie at cosmo. distances
 → Assumption of Euc. geom. breaks down
 loc) looking at younger gal: $\Rightarrow \Rightarrow \Rightarrow$ gal hasn't evolved to correct shape.

Def: Galaxy luminosity function: $\phi(L)$

$\phi(L)$: # of galaxies per unit volume per unit L

$$[\phi(L)] = \frac{N_g}{V L}$$

* In a range $L \pm dL$

$$\phi(L) dL : \frac{N_g}{Vol} : \frac{\# \text{ gal}}{Vol}$$

$$F_{lim} = \frac{L}{4\pi D_{max}^2} \Rightarrow D_{max}(L) = \left(\frac{L}{4\pi F_{max}} \right)^{1/2}$$

* Maximum volume sampled: (at a given L)

$$V_{max}(L) = \left(\frac{\Omega}{4\pi} \right) \left(\frac{4\pi}{3} D_{max}^3(L) \right)$$

$$V_{max}(L) = \frac{\Omega}{3} \left(\frac{L}{4\pi F_{max}} \right)^{3/2}$$

$$\bullet N(>F_{max}, L) dL = \left(\frac{\Omega}{3(4\pi)^{3/2}} \right) L^{3/2} F_{max}^{-3/2} \phi(L) dL$$

$$N(>F_{max}) = \left(\int_0^\infty L^{3/2} \phi(L) dL \right) \left(\frac{\Omega}{3(4\pi)^{3/2}} \right) F_{max}^{-3/2}$$

$N \propto F^{-3/2}$: regardless of $\phi(L)$

Intrinsic Gal. Lumin. func:

Observed dist. lum: $N(L)$
intrinsic dist. of lum $\phi(L)$

→ Different because of the volume-sampling eff (Malmquist Bias).

* In a flux limited survey, intrinsc L gal. will be over rep: they are visible to larger distances

$$[\phi] \sim \frac{N}{Vol L}$$

$$\phi dL \sim \frac{N}{Vol} dL$$

$$N(L) dL = V_{max}(L) \phi(L) dL$$

$$\phi(L) = \frac{N(L)}{V_{max}(L)}$$

Observed lum. fun:

$$\phi(L) = \left(\frac{\phi_*}{L_*} \right) \left(\frac{L}{L_*} \right)^\alpha e^{-L/L_*} \quad (\text{Schecter function})$$

L_* : characterstc lum above which the dist. decreases exp.

α : power-law slope which dominates at the faint-end

ϕ_* : Normalization

→ # of galaxies brighter than L_* decreases ex.

$$\alpha = -1.07$$

$$L_B^* = 24 \times 10^{10} L_\odot \quad (M_B^* = -20.5)$$

$$\phi^* = 0.047 \text{ Mpc}^{-3}$$

• Depends on morphology type:

• Bright end dominated by ellipticals

• Faint: late type star forming galaxies

ex: $\phi(L) = \frac{1}{L_0} L^\alpha e^{-L/L_0}$

$$\phi\left(\frac{1}{10}L_0\right) = \frac{\phi^*}{L_0} \left(\frac{1}{10}\right)^\alpha e^{-1/10} \sim 10/e^{1/10}$$

$$\phi\left(\frac{1}{2}L_0\right) = \frac{\phi^*}{L_0} \left(\frac{1}{2}\right)^\alpha e^{-1/2} \sim 2/\sqrt{e}$$

$$\phi(L_0) = \phi^*/L_0 e^{-1} \sim e^{-1}$$

$$\phi(2L_0) = \phi^*/L_0 \left(\frac{1}{2}\right)^\alpha e^{-2} \sim \frac{1}{2}e^{-2}$$

$$\phi(10L_0) = \phi^*/L_0 (10)^{-\alpha} e^{-10} \sim \frac{1}{10}e^{-10}$$

* Cosmic density of the universe:

$$\int_L^\infty \phi(L) dL = \phi^* \int_L^\infty \frac{1}{L_0} \left(\frac{L}{L_0}\right)^\alpha e^{-L/L_0} dL$$

$$= \phi^* \int_L^\infty \left(\frac{L'}{L_0}\right)^\alpha e^{-L'/L_0} d\left(\frac{L'}{L_0}\right)$$

$$u = \frac{L'}{L_0}$$

$$= \phi^* \int_{L/L_0}^\infty u^\alpha e^{-u} du$$

$$\left[\begin{aligned} \Gamma(n+1) &= n! = \int_0^\infty t^n e^{-t} dt \\ \Gamma((n+1), \beta) &= \int_\beta^\infty t^n e^{-t} dt \end{aligned} \right.$$

$$= \phi^* \Gamma(\alpha+1, \beta)$$

→ Increases w/o bound for $\alpha \geq -1$ as $L \rightarrow 0$

$$n_{gal} = \phi^* \Gamma(\alpha+1, L/L_0)$$

(num. den of galaxies)

Luminosity density:

$$\begin{aligned}\int_0^\infty L \phi(L) dL &= \int_0^\infty L \left(\frac{\phi_*}{L_*} \right) \left(\frac{L}{L_*} \right)^\alpha e^{-L/L_*} dL \\ &= \phi_* L_* \int_0^\infty \left(\frac{L}{L_*} \right)^{\alpha+1} e^{-L/L_*} d\left(\frac{L}{L_*} \right) \\ &= \phi_* L_* \Gamma(\alpha+2, L/L_*)\end{aligned}$$

$$n_L = \phi_* L_* \Gamma(\alpha+2, \frac{L}{L_*}) \quad (\text{int. lum density})$$

- diverges as $L \rightarrow 0$ and $\alpha \leq -2$
 $-2 < \alpha < -1$
- Bulk of galaxies comes from $\approx L_*$ galaxies

optical: $n_{L, \text{opt}} \approx 1.5 \times 10^8 L_\odot \text{ Mpc}^{-3}$

IR: $(n_L)_{\text{IR}} \approx 4.0 \times 10^8 L_\odot \text{ Mpc}^{-3}$

Mass densities:

- Assume a mean $\langle \frac{M}{L} \rangle$ across different galaxies we observe.

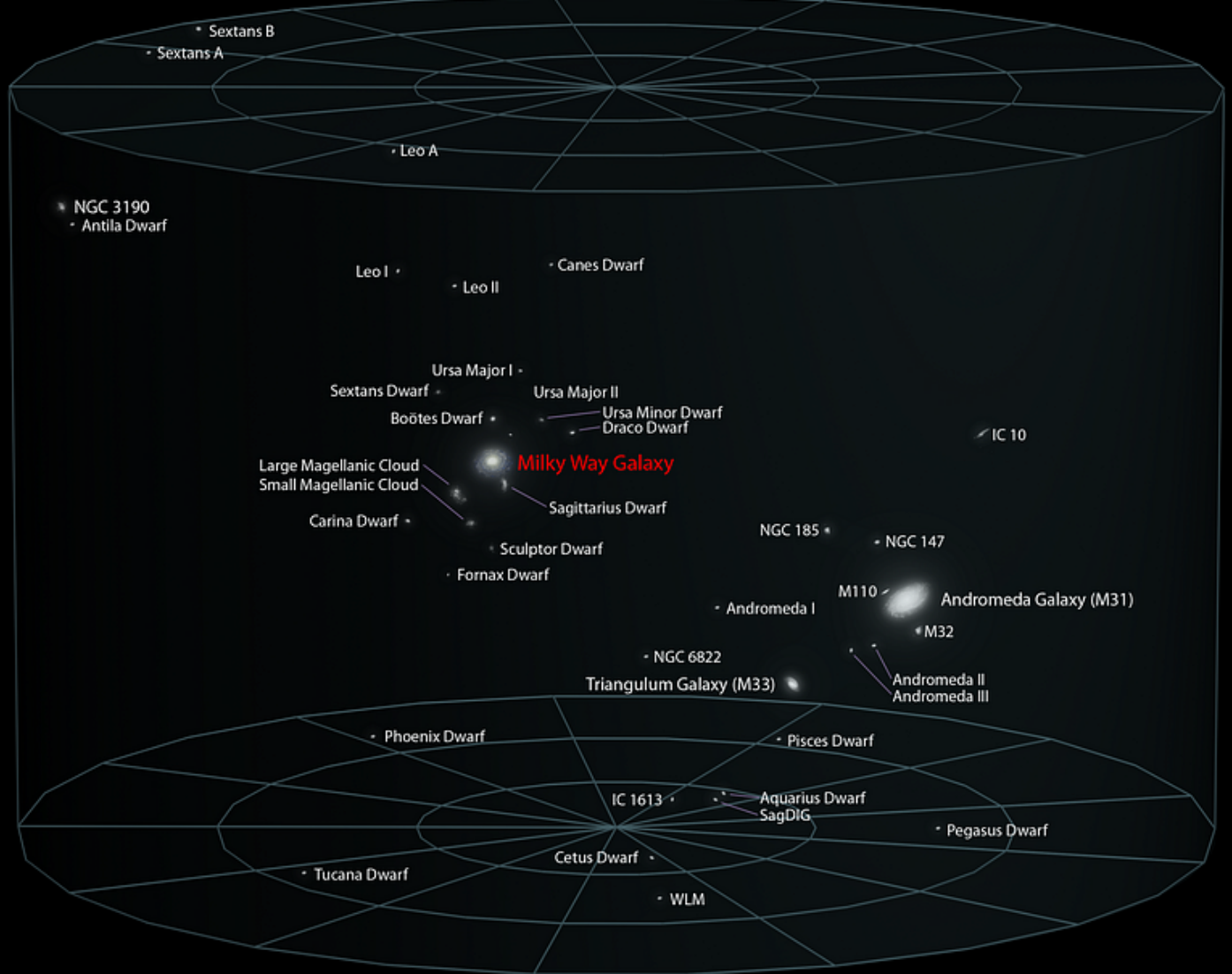
$$\langle \frac{M}{L} \rangle \sim 2 \text{ (stellar)} \quad \rho_{\text{mass}} \approx 6 \times 10^8 M_\odot \text{ Mpc}^{-3}$$

$$\langle \frac{M}{L} \rangle \sim 20 \text{ (M + D.M)} \quad \rho_{\text{mass}} \approx 6 \times 10^9 M_\odot \text{ Mpc}^{-3}$$

* Large scale spatial dist. of galaxies:

The local Group

Local Galactic Group



Local group: $D \approx 1 \text{ Mpc}$
 • 40 galaxies

- MW & M31 dominate group in $M + L$
- Gravitationally Bound
- Irregs + Dwarfs

• Pec velocities in group are neg: inwards

Neighbour Group: $D \approx 5 - 15 \text{ Mpc}$

- Brightest gal in our neighbourhood
- Bunch of other "local groups"

Virgo Cluster: $D \approx 20 - 100 \text{ Mpc}$

- 16 Mpc from us
- $C \rightarrow \sim 250$ large gal.
- ~ 2000 smaller

• Two main clumps

$V_r \approx 1200 \pm 500 \text{ km/s}$

• Sits in the middle of the local Supercluster

$\longleftrightarrow$
30 Mpc

• Zone of avoidance: large A_λ

• Super galactic plane

* Cornell Cluster: $\approx 100 \text{ Mpc}$

* $\geq 100 \text{ Mpc}$:

- redshift surveys
- 200 Mpc $\approx$ CZ

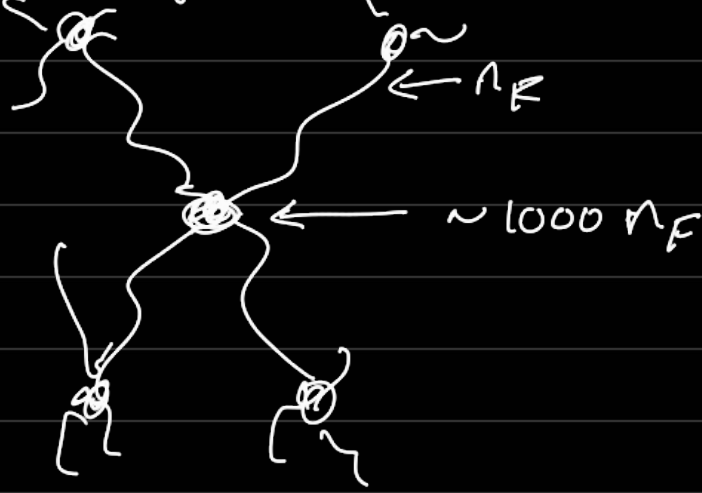
* The great wall: spans 70 Mpc

- Shows gal. dist. is not random
- Web structure occurs
- Empty voids $\sim 35 \text{ Mpc}$
- "filled" $\sim 70 \text{ Mpc}$
- Large clusters in 'convergent' point of galaxies

* Scales larger than 140: universe looks homogeneous & isotropic (define later)

* Clusters of galaxies:

- Most grav. bound structures in the universe



Regular clusters:

- Spherically symmetric
- 1000's of bright gal.
- Centrally conc.
- Very few spirals
- Coma: ($\approx 100 \text{ Mpc}$)

Irregular clusters:

- Irr. shape
- 100's bright gal.
- Loosely concentrated
- More spirals
- Virgo ($\approx 16 \text{ Mpc}$)

Morphology:

- Generally dominated by a few central ell.
- Usually a single: CD (Cluster Dominant)
CD: have an extended halo $\approx 100 \text{ kpc}$
 $I(r) \sim r^{-1.5}$
- Multiple nuclei: possibly mergers

- Morphology-density relation: S's are further out.

Gal. Cluster Vel. Disp.:

$$\sigma_r^2 = V_{rms}^2 = \langle V_{gal}^2 - \langle V_{gal} \rangle^2 \rangle$$

V_{gal} : vel. of an ind. gal.

$\langle V_{gal} \rangle$: mean vel. of all gal.

$$\sigma_r \approx 350 - 1000 \text{ km/s}$$

$$\frac{1}{2} M_{gal} \langle V_{gal} \rangle^2 = \frac{1}{2} M_{gal} \underline{3\sigma_r^2}$$

$$\tilde{\sigma}_r^2 = 3\sigma_r^2 \Rightarrow \tilde{\sigma}_r = \sqrt{3}\sigma_r : 600 - 1700 \text{ km/s}$$

Intergalactic Gas in Gal. Clusters:

- Diffuse X-ray emissions in cluster 'open' areas

Intra-Cluster Medium (ICM)

- Brightest X-ray sources aside from AGN

- Gas has $T \approx 10^7 - 8 \text{ K}$

→ Produced by Bremsstrahlung emission

- Fairly well dist

$$\frac{3}{2} kT = \frac{1}{2} m v^2 : \quad m_p \approx 10^{-27} \text{ kg}$$

$$k_B \approx 10^{-23} \text{ J/K}$$

$$3 \cdot 10^7 \cdot 10^{-23} \cdot 10^{27} = v^2$$

$$3 \times 10^3 = v \approx 547 \text{ km/s}$$

- Single proton: m_p

→ Gas in a potential well

$$U = - \frac{G M_{clust} m_p}{R_{clust}}$$

- Virialised cluster

$$2K + U$$

$$K = - \frac{1}{2} U$$

$$K = \frac{3}{2} k T_{\text{clust}}$$

$$\frac{3}{2} k_B T_{\text{clust}} = \frac{1}{2} \frac{G M_{\text{cluster}} m_p}{R_{\text{cluster}}}$$

$$T_{\text{cluster}} \approx \frac{G M_{\text{cluster}} m_p}{3 k_B R_{\text{cluster}}}$$

• electrons don't gain much en. from virialization
 $m_e \ll m_p$

• Recall: half goes into heat, other into radiation, so el. get half from equilibrium

$$T_{\text{clust}} = \frac{G M_{\text{cluster}} m_p}{6 k_B R_{\text{cluster}}}$$

ex $M \approx 10^{15} M_{\odot}$, $R \approx 3 \text{ Mpc} \Rightarrow T \approx 3 \times 10^7 \text{ K}$

Origin of ICM gas:

- Continuum emission
- Fe lines are prominent: indicate a heavy element content $\sim 0.3 - 0.5 Z_{\odot}$

ICM is not Primordial: been through many mixtures

(i) Ram pressure stripping

$$P_{\text{ram}} \approx \rho_{\text{ICM}} V^2$$

V : infall vel.

$$F = \frac{dP}{dt} = \frac{\Delta M}{\Delta t} V$$



$$\frac{\Delta M}{\Delta A \Delta t V} = \rho$$

$$\frac{\Delta M}{\Delta t} = \Delta A V \rho_{\text{ICM}}$$

$$\frac{\frac{\Delta E}{\Delta t}}{\Delta t} = P_{\text{ram}} = \rho_{\text{ram}} v^2$$

* Galaxy cluster masses: $2K + U = 0$

$$U = -\alpha \frac{GM^2}{R} \quad (\text{looked at AGNs})$$

$\alpha \approx 1$ for Gal. clusters

$$K = \frac{3}{2} M_{\text{clust}} \sigma_r^2 \quad \sigma_r: \text{line-of-sight vel. disp.}$$

$$3 M_{\text{clust}} \sigma_r^2 \approx \alpha \frac{GM_{\text{clust}}^2}{R_{\text{clust.}}}$$

$$M_{\text{clust.}} = \frac{3 \sigma_r^2 R_{\text{clust.}}}{\alpha G}$$

For $R \approx 3 \text{ Mpc}$, $\sigma_r \approx 1000 \text{ km s}^{-1}$
 $M_{\text{clust}} \approx 10^{15} M_{\odot}$

- Want to relate this to stellar mass we see in galaxies
- Use $[M/L]$
- Observations of $L \approx 10^{12-13} L_{\odot}$
- $\left[\frac{M}{L}\right] \approx 2$ (stellar)

$$\left[\frac{M}{L}\right] \approx \frac{10^{15} M_{\odot}}{10^{13} L_{\odot}} \approx 50 \sim 500 \left(\frac{M_{\odot}}{L_{\odot}}\right)$$

→ There is not enough stellar (radiative) matter to account for our observed values

* Even when we use $\left[\frac{M}{L}\right] \approx 10^{-20}$
(M + D.M.)

→ Individual galaxies can only account for
(20) (5%) of the total mass in clusters

→ D.M. (Next time)

Hot Gas studies:

$$M_{cl} = \frac{6 k_B T_e R}{G m_p}$$

T_e : X-ray spectra

• Use



$$R \approx D \theta$$

Mass of the ICM:

• Emission/vol:

$$\epsilon_{ff} = 1.48 \times 10^{-40} n_e T^{1/2} g_{ff} \text{ W m}^{-3}$$

g_{ff} : Gaunt factor ≈ 1.2

$$L_x = \epsilon_{ff} \frac{4\pi}{3} R^3 = R^3 \cdot 7.4 \times 10^{-40} n_e^2 T^{1/2} \text{ W}$$

R : from flux & θ

T : X-ray spectrum

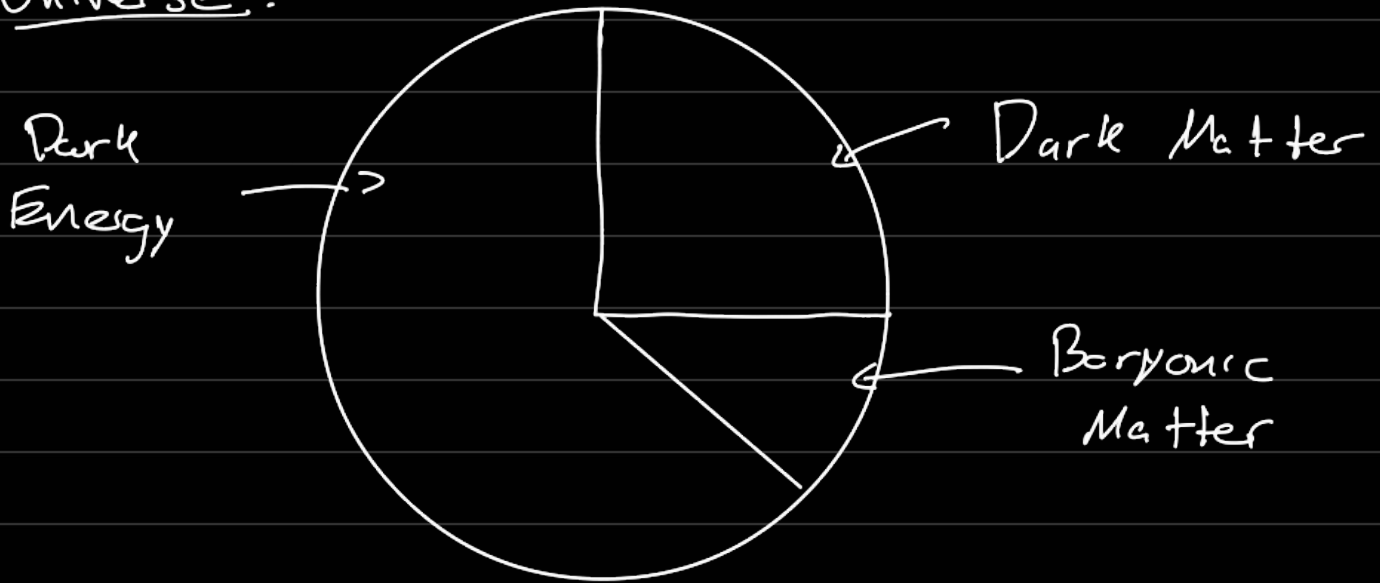
find n_e from here

$$M_{gas} = \frac{4\pi}{3} R^3 n_e m_p$$

- $M_{gas} \approx 0.1 - 0.2 M_{cluster} (\approx 10^{14} M_{\odot})$
still not dark matter.

* CMB, gal. clus., redshifts, Type Ia super
→ Λ CDM (Cold Dark Matter)

Universe:



CMB: Surface of last scattering $z = 1100$
• Tells us initial conditions of obs. universe

Web: filaments, clusters, voids

→ Dark Matter Halo function:

$$\frac{\# \text{ of D.M. halos}}{(\text{Vol})(\text{mass})}$$

• Simple model: 1 halo (D.M.) $\leftrightarrow$ 1 gal.

M_H : Mass of halo

• has $f_b \times M_H$ baryons
 $f_b \approx 0.16$ Baryon fraction $\left(\frac{\Omega_b}{\Omega_M} \right)$

Galaxy Stellar Mass function:

* Star forming eff. peaks at $\approx 20\%$.

→ Feedback: Supernova feedback causes no low mass gas

S.N: $\approx 10^{45}$ J in IGM

• Comparable to binding energy of a small gal.
Sitting in a low mass dark matter halo

$$E_{\text{Tot}} = K + U = -K$$

$$E_{\text{bind}} = K \approx \frac{1}{2} M_{\text{gal}} \sigma^2 \approx E_{\text{S.N}}$$

AGN:

• Galaxy evol. models involve AGN feedback

• Binding $E_{\text{bind}} \approx \frac{1}{2} M_{\text{gal}} \sigma^2$

• En. of Supermass. B.H.s

→ 10% acc. goes to Radiation
 $\epsilon \approx 0.1$ $E = \epsilon M c^2$

→ 90% grows the B.H.

$$E_{\text{bh}} \approx 0.1 M_{\text{ph}} c^2$$

↑ Final mass

$$E_{\text{bh}} > 200 E_{\text{bind}}$$

(i) Quasar mode

(ii) Radio mode ← Radio jets

→ prevents gas cooling / star forming regions

Star forming rates:

Kennicutt-Schmidt Law

$$\Sigma_{\text{SFR}} \propto \Sigma_{\text{gas}}^{1.4 \pm 0.15} \quad (\text{SFR})$$

Σ_{gas} : surface density of H

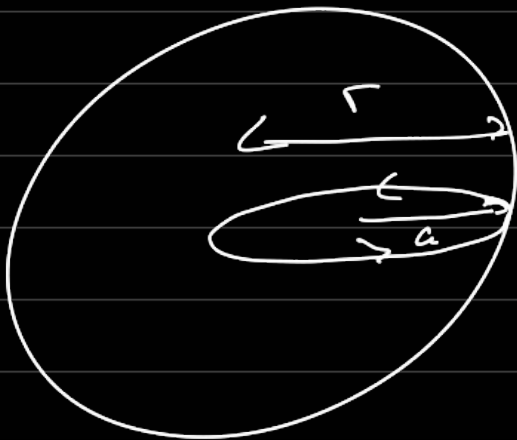
$$\text{SFR} \propto \frac{\rho_{\text{gas}}}{t_{\text{ff}}}$$

FF: Kepler's 3rd:

$$\frac{P^2}{a^3} = \frac{4\pi}{GM}$$

P: period
a: semi major

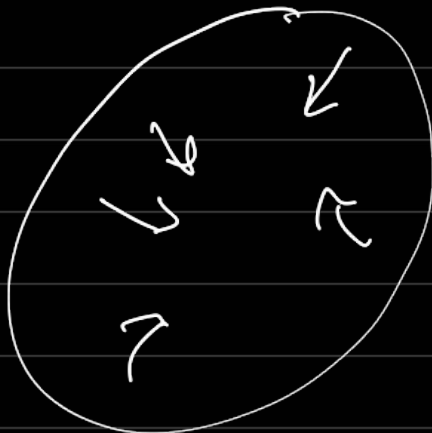
t_{ff} : time needed for the cloud to fully collapse



$$a = \frac{r}{2} \quad | \quad t_{\text{ff}} = \frac{P}{2}$$

$$M = \frac{4\pi}{3} r^3 \bar{\rho}$$

• Same as star
grav > pressure



$$(\text{SFR}) \propto \rho_{\text{gas}}^{1.5}$$

* Star form at $z > 1$:

• Lyman break gal.

→ absorption of neutral hydrogen

$$(1+z) \pm 2/6 \text{ \AA}$$

$$(1+z) 9/2 \text{ \AA}$$

$$\rho_{\nu}(z) = \int_0^z \phi_{\text{SFH}}(z) \frac{dz}{dz} dz$$

